What have you learned thus far?

- What is
  - Distance measure or Gaussian Kernel?
  - Isolation Kernel?
  - Isolation Distributional Kernel (IDK)?
- How does IDK differ from Gaussian Distributional Kernel?
- Describe the three different kinds of clustering algorithms, and their strengths and weaknesses.

Data Mining Algorithms

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Required Reading

- Aggarwal Chapters 4 (up to 4.4.3.1), 6 (6.1, 6.3,6.6 only), 8 (8.1,8.5,8.6 only), 10 (10.1, 10.3-10.4, 10.6 only)
- F. T. Liu, K. M. Ting, and Z.-H. Zhou. Isolation forest. *IEEE ICDM Conference*, pp. 413–422, 2008.
- K. M. Ting, B.-C. Xu, T. Washio, Z.-H. Zhou (2020) Isolation Distributional Kernel: A new tool for kernel based anomaly detection, Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, pp. 198–206.

Acknowledgement:

Unless stated otherwise, the contents and images on the following slides are sourced from

C. C. Aggarwal, Data Mining: The Textbook. Springer, 2016.

Contents

1. Introduction
2. Classification
3. Clustering
4. Anomaly Detection
5. Association pattern mining

Introduction

Classification: Problem definition

Definition 1 (Data Classification) *Given an $n \times d$ training data matrix D (database D), and a class label value in $\{1 \ldots k\}$ associated with each of the n rows in D (records in D), create a training model M, which can be used to predict the class label of a d-dimensional record $Y \notin D$.*

Classification algorithms

- Decision Trees
- Rule-based classifier
- K nearest neighbor classifier
- Support Vector Machines
- Neural Networks

Algorithm for Decision Tree Construction

Basic algorithm (a greedy algorithm):

Tree is constructed in a top-down, recursive, divide-and-conquer manner

- Start with a set of training data
- If all examples belong to the same class

  - Make a leaf labeled with that class

- Otherwise choose a test based on one attribute—each test attribute is selected on the basis of a heuristic or statistical measure (e.g. information gain)

  (This becomes an internal node); then create

  - One branch for each value of categorical attribute, or
  - Two branches: A<=T, A>T for numeric attribute A, threshold T.

  Subdivide the examples according to their outcome on the test.

- Apply the procedure recursively to each subset.

Algorithm for Decision Tree Construction

Conditions for stopping partitioning

- All data in a given node belong to the same class
  - the given node then becomes a leaf node
- There are no more remaining attributes for further partitioning
- There are no more data left to do further partitioning

Finding the Split

- An attribute has to be chosen to split the data while building the tree
- This chosen attribute does the best job of discriminating data among their classes, according to some criterion.
- One approach
    - choose the attribute which minimises the diversity/impurity amongst its children nodes
- Several criteria have been proposed
    - Information Gain/Entropy, Gini, Chi-Squared, Rough Sets etc.

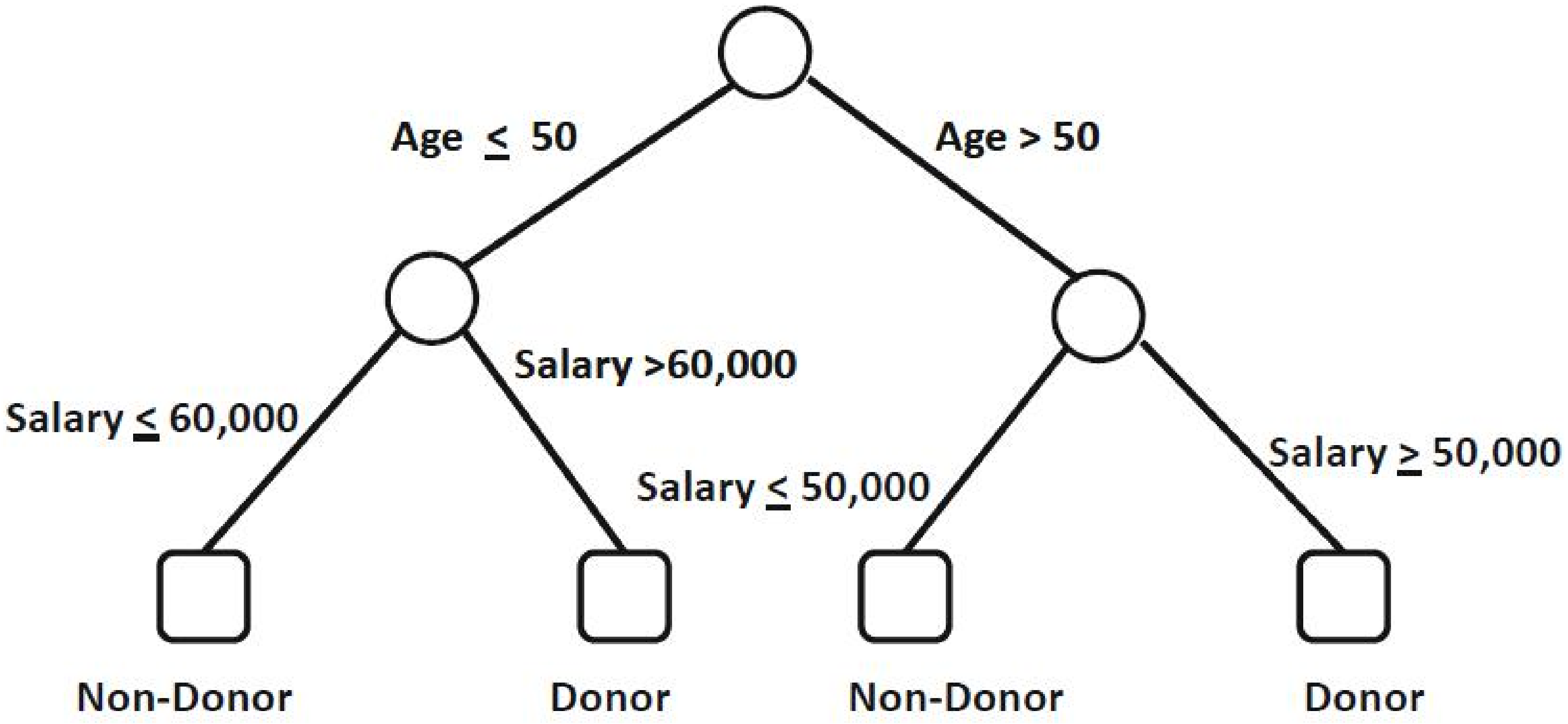
How to select the attribute for splitting at each internal node?

Information gain (C4.5 or J48 in WEKA)

- Gain is the difference between the amount of information that is needed to make a correct prediction before a split has been made, and that required after the split
- If the amount of information required is lower after the split is made, then the split is said to have decreased the class-mix (impurity) of the original data

Algorithm *GenericDecisionTree*(Data Set: $\mathcal{D}$)
begin
  Create root node containing $\mathcal{D}$;
  repeat
    Select an eligible node in the tree;
    Split the selected node into two or more nodes
      based on a pre-defined split criterion;
  until no more eligible nodes for split;
  Prune overfitting nodes from tree;
  Label each leaf node with its dominant class;
end

Example



Name	Donor?
Nancy	N
Jim	N
Jane	N
Steve	N

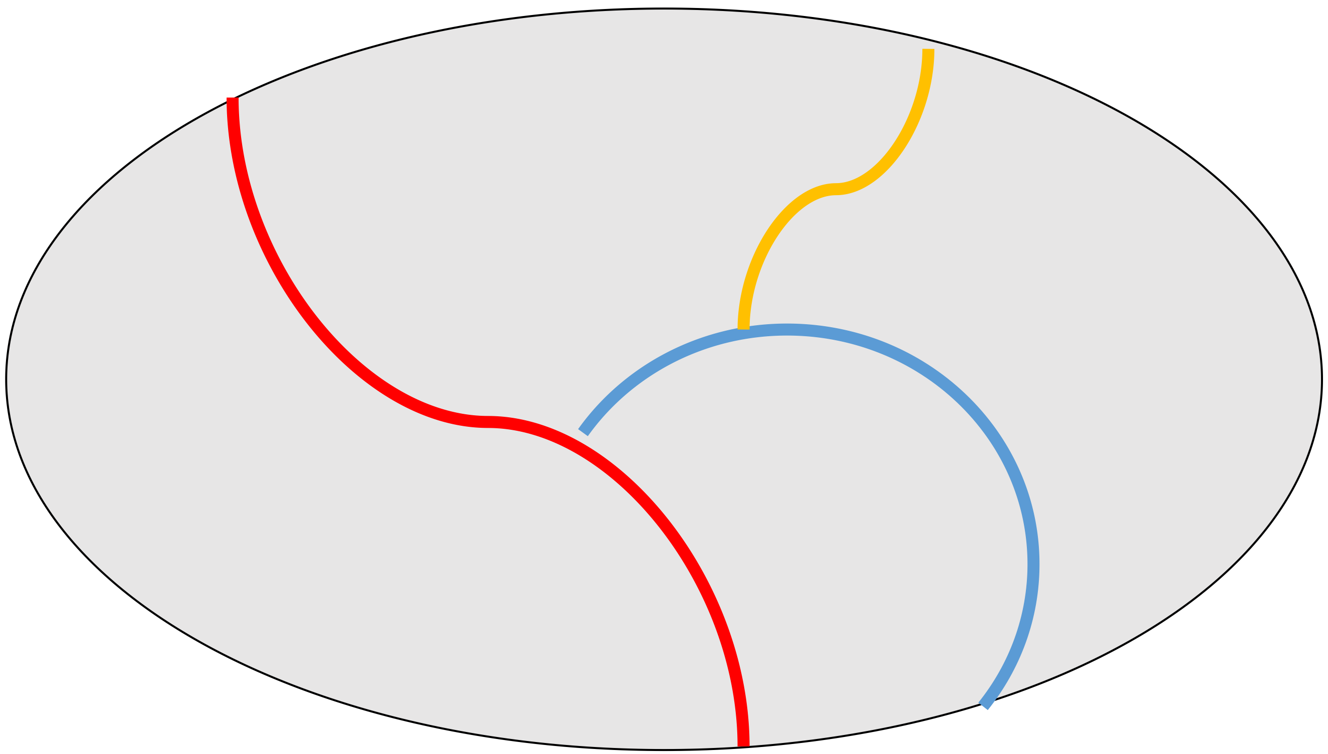
Name	Donor?
Allen	Y

Name	Donor?
Mary	N

Name	Donor?
Peter	Y
Sayani	Y
Lata	Y
Victor	Y
Dale	Y

Set-covering (sequential covering) Algorithm

- It is used to learn a set of decision rules

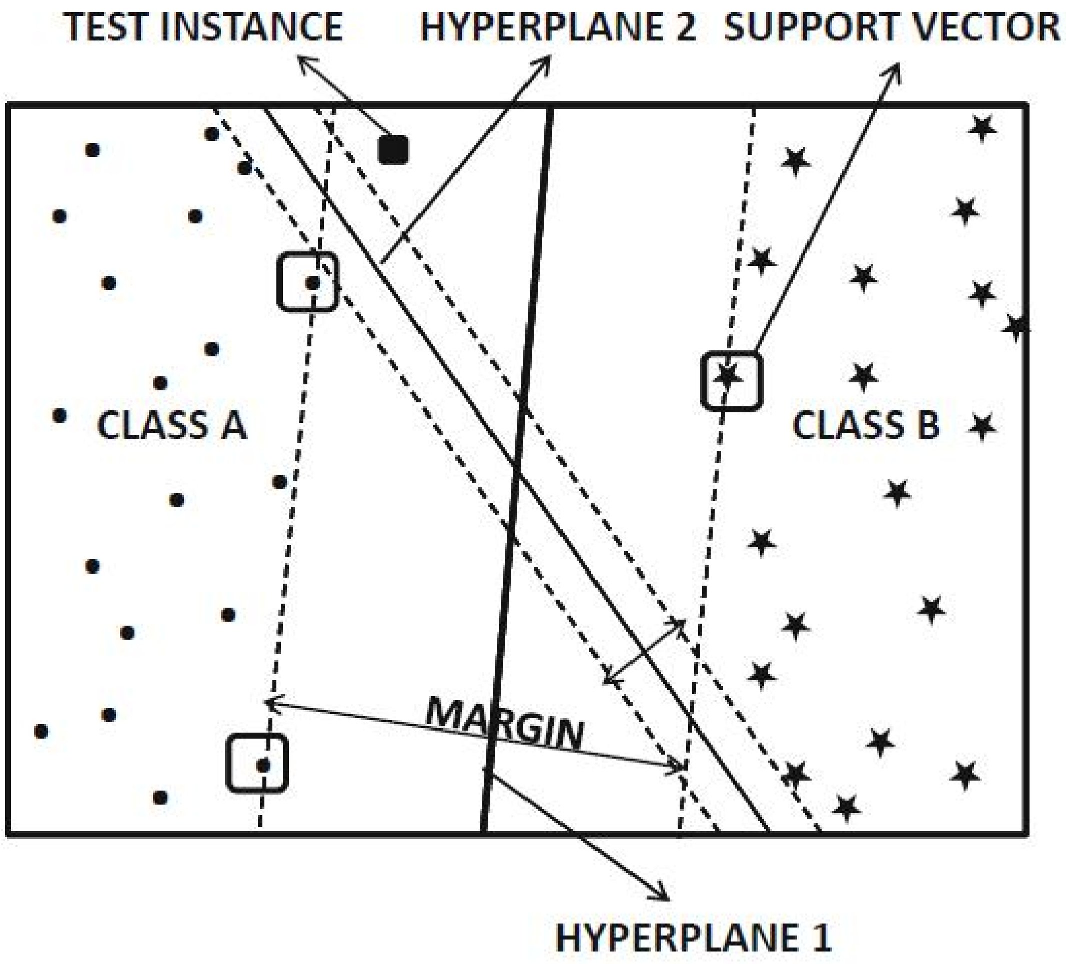


K-Nearest Neighbours

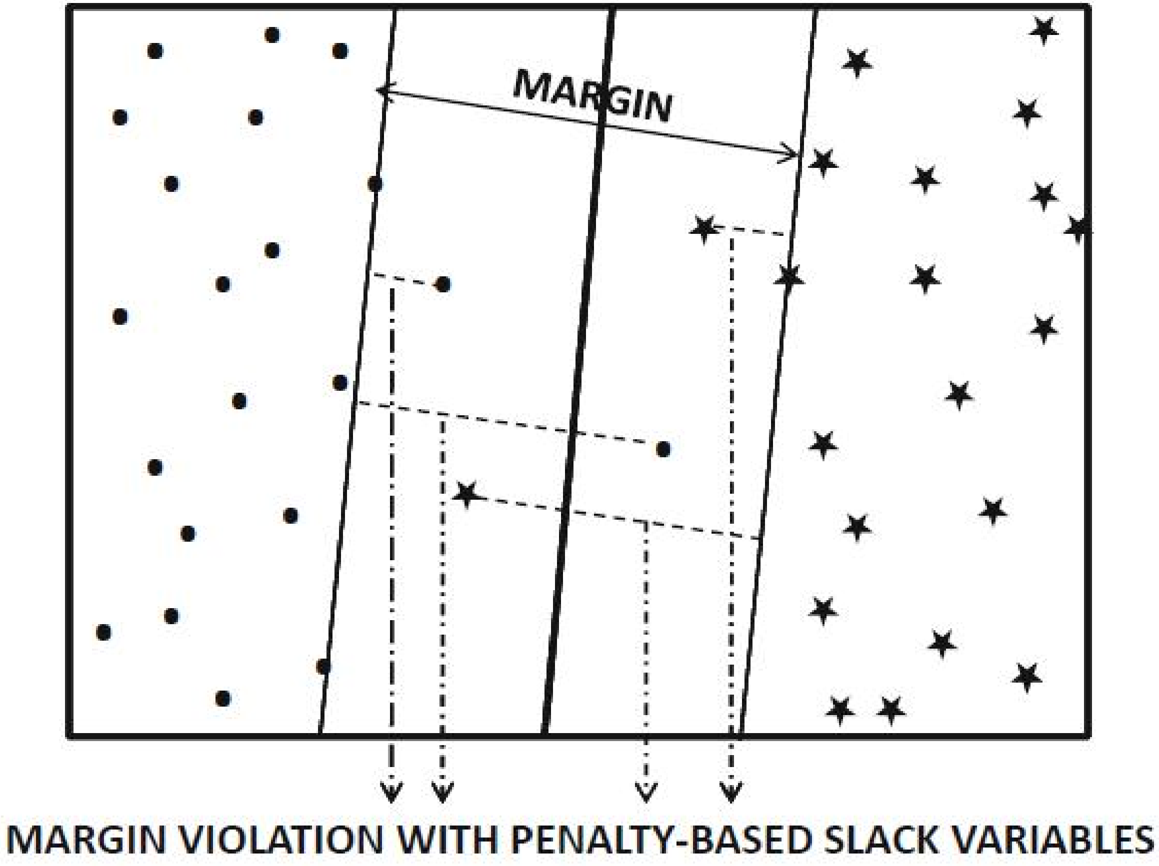
- Instance-based learning algorithms
- Lazy learning algorithms
- Represent the model in its raw form i.e., the given training examples---No work is conducted during the training process.
- The following functions are required:
  - Similarity function (need normalisation)
  - Decision function (k needs to be specified by user)
  - Improve search through indexing

Support Vector Machines for Linearly Separable Data

Based on the minimum structural risk principle: maximizing margin



(a) Hard separation



(b) Soft separation

Support vectors

- Boundary: $y_j(W \cdot X_j + b) \geq \ +1$
- The weight vector $W = \sum_{i=1}^{n} \lambda_i y_i X_i$
- With s support vectors, the boundary can be expressed as

$$y_j(\sum_{i=1}^{s} \lambda_i y_i X_i \cdot X_j + b) \geq \ +1$$

- The dot product enables the use of a kernel via kernel trick, i.e., $k(X_i, X_j) = \phi(X_i) \cdot \phi(X_j)$

Nonlinear Support Vector Machines

- The kernel trick leverages the important observation that the SVM formulation can be fully solved in terms of dot products (or similarities) between pairs of data points. One does not need to know the feature values $\phi(X)$. Therefore, the key is to define the pairwise dot product (or similarity function) in the *d*-dimensional transformed representation $\phi(X)$, with the use of a kernel function $k(X_i, X_j)$.

$$k(X_i, X_j) = \phi(X_i) \cdot \phi(X_j)$$

- The kernel trick can be viewed as a means to find linear boundary in high dimensional (Hilbert) space of $\phi(X)$. Gaussian or RBF kernel has infinite-dimensional $\phi(X)$.

Neural Networks

- A **neural network,** a simplified version of biological cells, consists of interconnected units called neurons that send signals to one another.
- Individual neurons are simple, but putting them in a network can perform complex tasks.
- In machine learning, an *artificial neural network* (ANN) is a model used to approximate nonlinear functions.
- A simple ANN has one hidden layers of neurons coupled with the input layer and output layer.
- A deep neural network has many hidden layers.

Reading guide

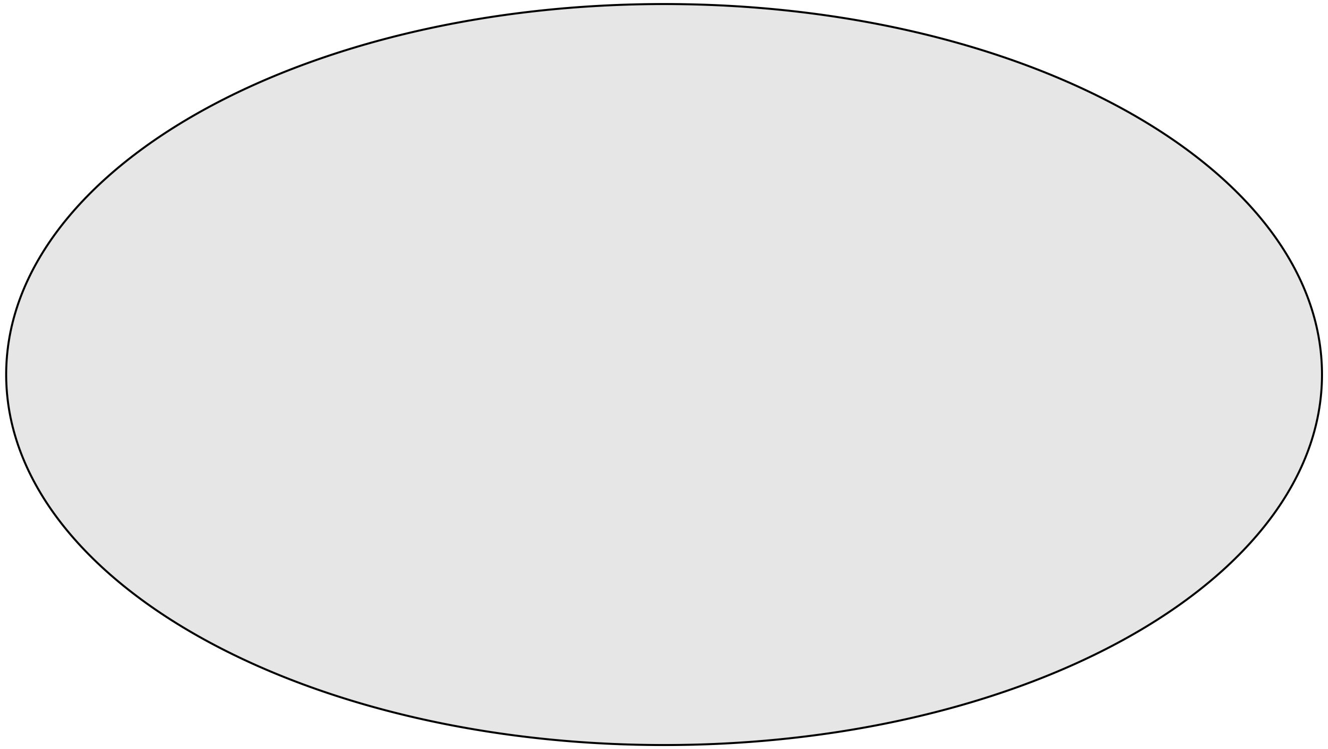
- Section 10.6 SVM: Focus on the key ideas of SVM: learning a linear model using kernel trick. You may skip the description on optimization as it is not our focus in this course.

Discussion #1

1. Explain the similarities and differences between the main classification methods.
2. Explain why the divide-and-conquer algorithm used to generate a decision rule is called a greedy algorithm.
3. The set-covering algorithm used to generate decision rules is sometimes called the separate-and-conquer algorithm. How does this differ conceptually from the divide-and-conquer algorithm?
4. As the k-nearest neighbor algorithm does nothing in training, does it produce a generalized model that enables it to classify well on unseen data points?
5. Explain the distinct features of SVM with respect to other classification methods.
6. Which of the five classification algorithms require a distance/similarity function?

Set-covering (sequential covering) Algorithm

- It is used to learn a set of decision rules



Clustering: Problem definition

Definition 2 (Data Clustering) *Given a data matrix D, partition its rows (records) into sets $C_1 \ldots C_k$, such that the rows (records) in each cluster are "similar" to one another.*

Clustering methods

- K-means/k-medoids
- DBSCAN
- Spectral clustering
- IDK-based Clustering (e.g., psKC)

Spectral Clustering (three basic steps)

1. Preprocessing
   - Produce a similarity matrix of a neighborhood graph

2. Eigen-decomposition
   - Compute eigenvalues & eigenvectors of the similarity matrix
     or Laplacian matrix
   - Map each point in the input space to the new space,
     represented by k eigenvectors

3. Clustering (often uses **k**-means)

Discussion #2

1. Describe the two key steps in k-means clustering.
2. Describe the two key steps in DBSCAN.
3. Explain the purpose of the first two steps in Spectral Clustering.
4. Explain the key difference between IDK-based Clustering and the above three clustering methods.

Outlier detection: Problem definition

Definition 3 (Outlier Detection) *Given a data matrix D, determine the* ***few*** *rows of the data matrix that are very* ***different*** *from the* ***majority*** *of rows in the matrix.*

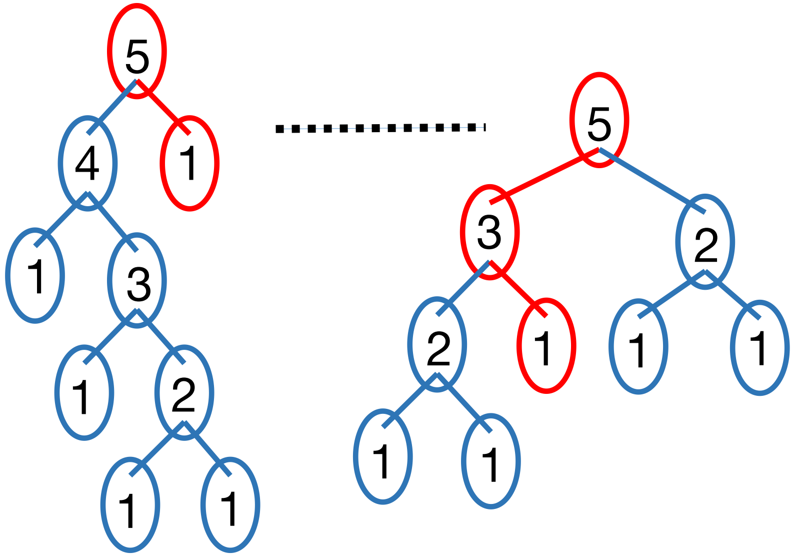
Other names: anomaly detection, novelty detection

Outlier detection methods

- Distance-based anomaly detectors
- Local Outlier Factor (LOF)
- One-class Support Vector Machine (OCSVM)
- Kernel Density Estimator (KDE)
- Isolation-based anomaly detectors:
  - Isolation Forest (iForest)
  - Isolation Distribution Kernel detector
    - Based on Kernel Mean Embedding

2. iForest (Liu et al ICDM 2008)

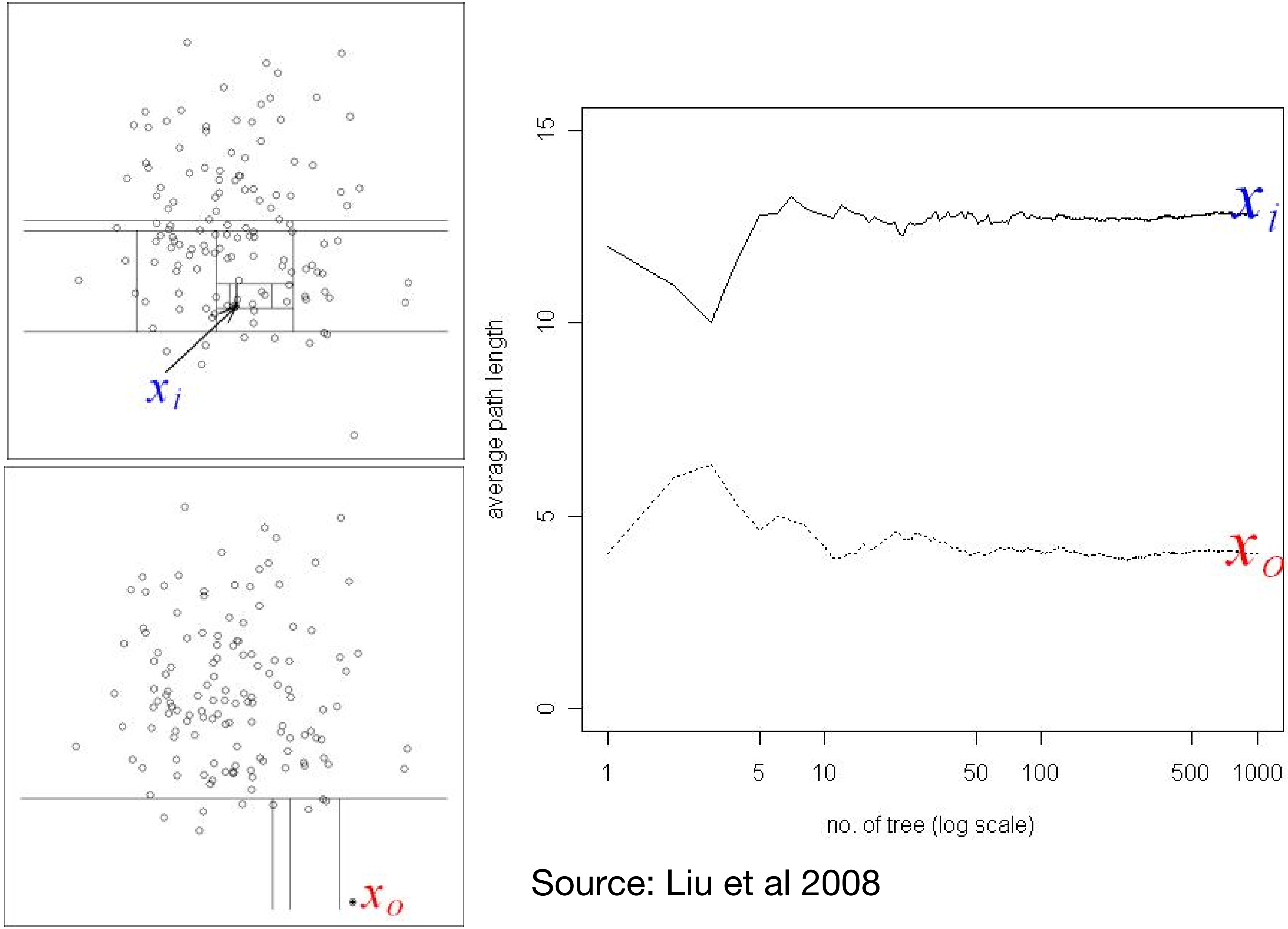
- A collection of isolation trees (iTrees)
- Each iTree isolates every instance from the rest of the instances in a given sample
- Anomalies are ‘few and different’
  - More susceptible to isolation
  - Shorter average path



$$Score(\boldsymbol{x}) = \frac{1}{t}\sum_{i=1}^{t} \ell_i(\boldsymbol{x})$$

where $\ell_i(\boldsymbol{x})$ is the path length of $\boldsymbol{x}$ traversed in tree i

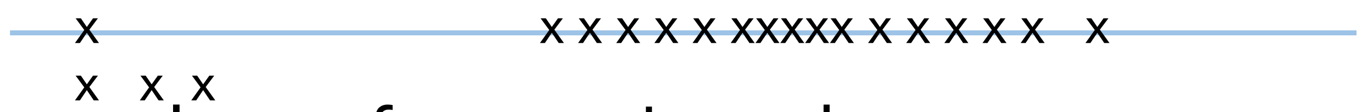
Isolation example



Source: Liu et al 2008

Construction of iTrees

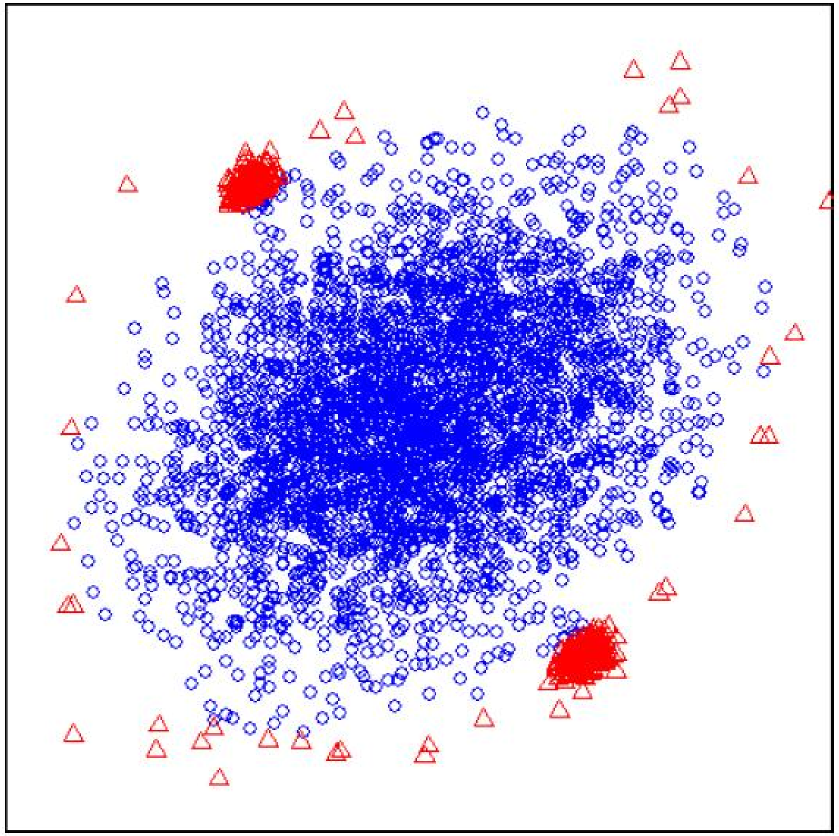
- iTrees are completely random trees
  - guided by the range of values in a subsample



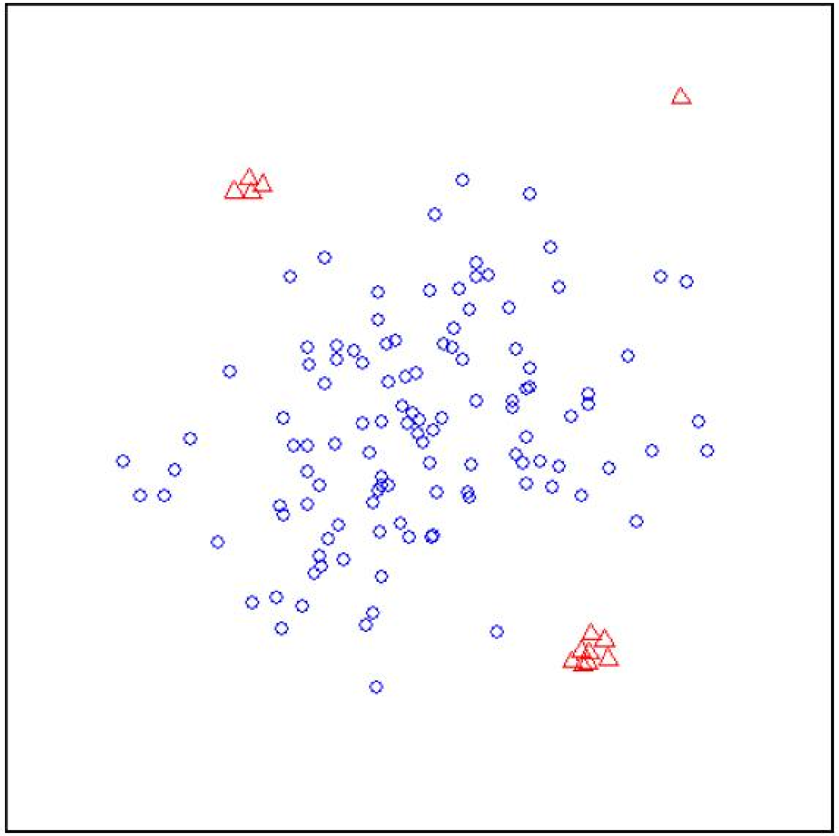
- Subsamples $\mathcal{D}_i$ from a given data set D
  - $\mathcal{D}_i \subset D \ (i = 1,2,\cdots t)$
  - $|\mathcal{D}_i| = \psi \ll |D| = n$
- Build an iTree from each $\mathcal{D}_i$
  - Partition recursively until every instance is isolated
  - Height restriction ($\mathrm{h} = \log_2 \psi$)

The effect of subsampling

- Reduce both the swamping and masking effects



(a) Original sample
(4096 instances)



(b) Sub-sample
(128 instances)

Source: Tony Liu's PhD Thesis
– Anomaly Detection by Isolatio

Time Complexity

	iForest	LOF
Model building	$O(t\psi\log(\psi))$	NA
Evaluation	$O(tn\log(\psi))$	$O(n^2)$

n – number of instances in the given data set

ψ – number of instances in a subsample used to build an iTree, where $\psi \ll n$

t – ensemble size

Background: Kernel Mean Embedding

- Kernel mean embedding (KME) [*] is a way to convert a point-to-point kernel to a distributional kernel

$$\widehat{K}(P_S, P_T) = \frac{1}{|S||T|}\sum_{x\in S}\sum_{y\in T}\kappa(x, y)$$

$$\approx \langle\widehat{\varphi}(P_S), \widehat{\varphi}(P_T)\rangle \qquad \text{(using Nystrom method)}$$

- If κ is a characteristic kernel, then its kernel mean map is injective, i.e., $\parallel \widehat{\varphi}(P_S) - \widehat{\varphi}(P_T)\parallel_H \ = 0 \ if \ and \ only \ if P_S = P_T$

- KME, which employs Gaussian kernel, has two key issues:
  - Quadratic time complexity (resolved using Nystrom method)
  - Weak task-specific accuracy

[*] Alex Smola, Arthur Gretton, Le Song, and Bernhard Schölkopf. 2007. A Hilbert Space Embedding for Distributions. In Algorithmic Learning Theory, 13–31.

Kernel Mean Embedding: root cause of key issues

- KME, which employs Gaussian kernel, has two key issues:
  - Quadratic time complexity (resolved using Nystrom method)
  - Weak task-specific accuracy
- The root cause of these issues: Gaussian kernel

$$\widehat{K}(P_S, P_T) = \frac{1}{|S||T|} \sum_{x \in S} \sum_{y \in T} \kappa(x, y)$$

- Replace Gaussian kernel with Isolation Kernel to produce Isolation Distribution Kernel (IDK) [*]

[*] K. M. Ting, B.-C. Xu, T. Washio, Z.-H. Zhou (2020) Isolation Distributional Kernel: A new tool for kernel based anomaly detection, KDD, pp. 198–206.

IDK method for anomaly detection

The idea is to use IDK to measure the similarity effectively between every point (in the given dataset *D*) and *D* (the reference dataset).

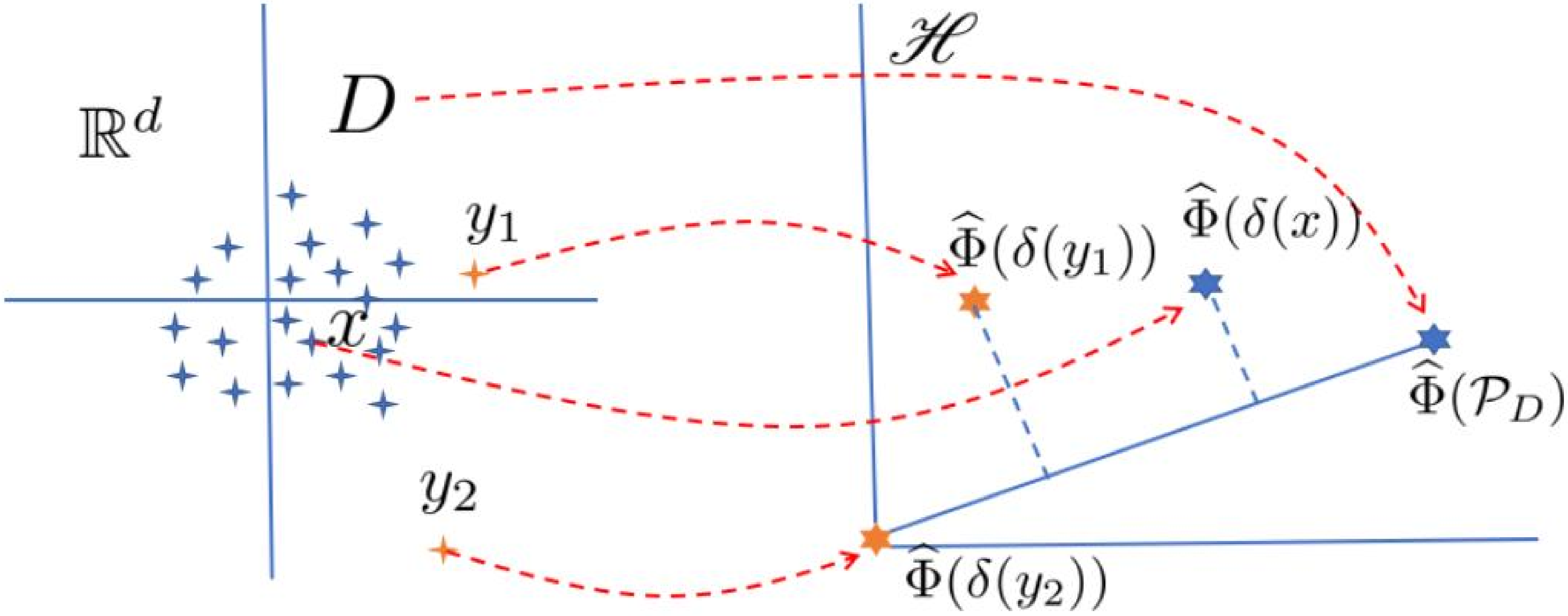
- If $x \sim \mathcal{P}_D$, $\widehat{\mathcal{K}}(\delta(x), \mathcal{P}_D)$ is large, which can be interpreted as x is likely to be part of $\mathcal{P}_D$.
- If $y \nsim \mathcal{P}_D$, $\widehat{\mathcal{K}}(\delta(y), \mathcal{P}_D)$ is small, which can be interpreted as y is not likely to be part of $\mathcal{P}_D$.

Definition of anomaly:

'Given a similarity measure $\widehat{\mathcal{K}}$ of two distributions, an anomaly is an observation whose Dirac measure δ has a low similarity with the distribution from which a reference dataset is generated.'

Example computation

$$\widehat{\mathcal{K}}_I(\delta(x), \mathcal{P}_D) \quad = \quad \frac{1}{t}\left\langle \widehat{\Phi}(\delta(x)), \widehat{\Phi}(\mathcal{P}_D) \right\rangle$$



Discussion #3

1. Describe the operational principle of
   - Distance-based anomaly detector.
   - Isolation-based anomaly detector.
   - IDK-based anomaly detector.
2. Is an Isolation tree a variant of decision tree? Explain your answer.
3. Explain another possible way to perform isolation which is different from using isolation trees. Hint: the principle is to isolate one point from the rest of the points in a sample.

Frequent pattern mining: Problem Definition

Definition 4 (Frequent Pattern Mining) *Given a binary $n \times d$ data matrix D, determine all subsets of columns such that all the values in these columns take on the value of 1 for at least a fraction s of the rows in the matrix. The relative frequency of a pattern is referred to as its support. The fraction s is referred to as the minimum support.*

Definition 4.1 (Association Rules) *Let A and B be two sets of items. The rule $A \Rightarrow B$ is said to be valid at support level s and confidence level c, if the following two conditions are satisfied:*

1. The support of the item set A is at least s.

2. The confidence of $A \Rightarrow B$ is at least c.

Association Pattern Mining algorithms

- Enumeration-tree algorithms
- Apriori algorithm

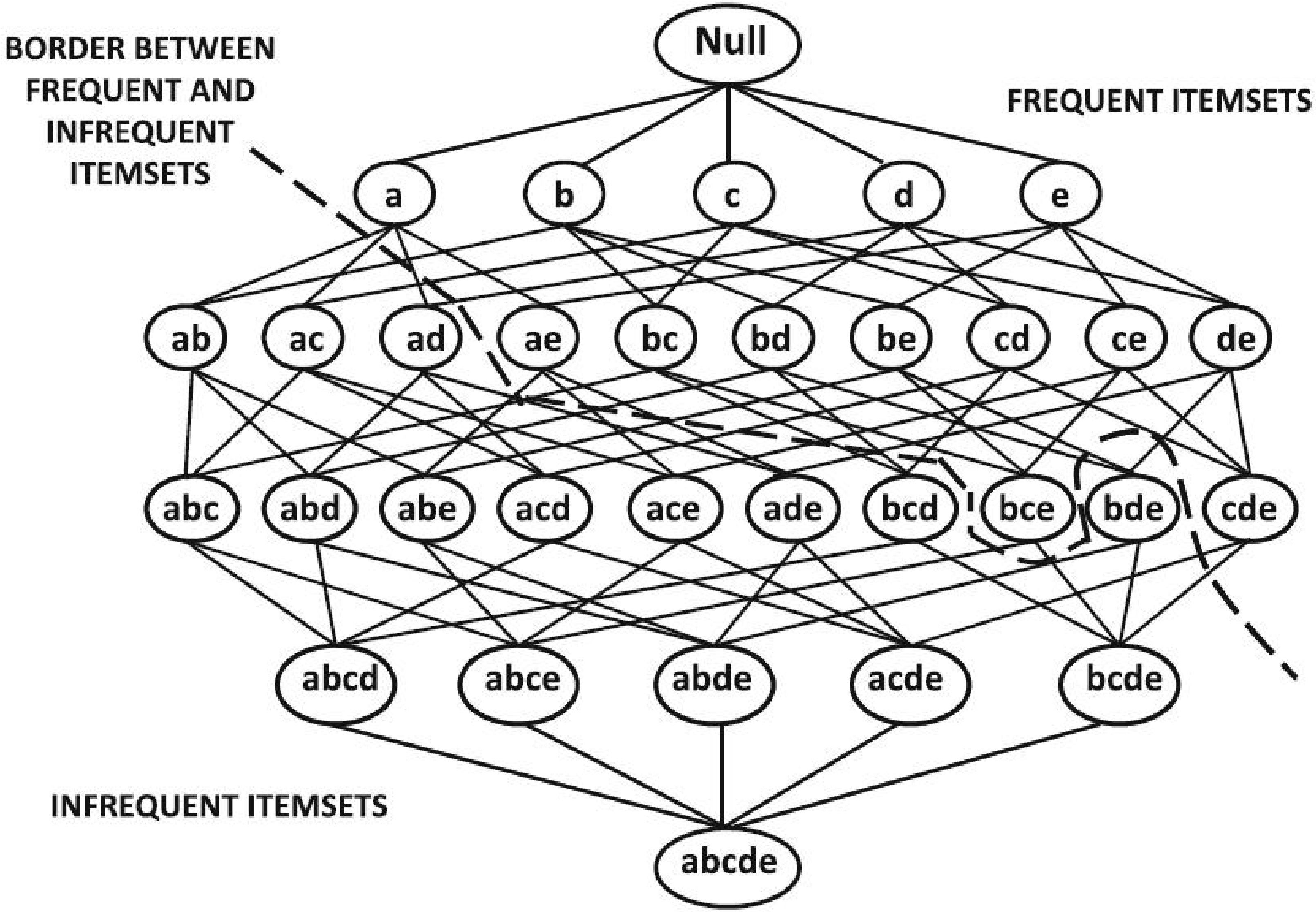


Figure 4.1: The itemset lattice

Algorithm $Apriori$(Transactions: $\mathcal{T}$, Minimum Support: $minsup$)
begin
  $k = 1$;
  $\mathcal{F}_1 = \{$ All Frequent 1-itemsets $\}$;
  while $\mathcal{F}_k$ is not empty **do begin**
    Generate $\mathcal{C}_{k+1}$ by joining itemset-pairs in $\mathcal{F}_k$;
    Prune itemsets from $\mathcal{C}_{k+1}$ that violate downward closure;
    Determine $\mathcal{F}_{k+1}$ by support counting on $(\mathcal{C}_{k+1}, \mathcal{T})$ and retaining itemsets from $\mathcal{C}_{k+1}$ with support at least $minsup$;
    $k = k + 1$;
  end;
  return$(\cup_{i=1}^{k} \mathcal{F}_i)$;
end

Figure 4.2: The *Apriori* algorithm

Downward closure property:
Every subset of a frequent itemset is also frequent.

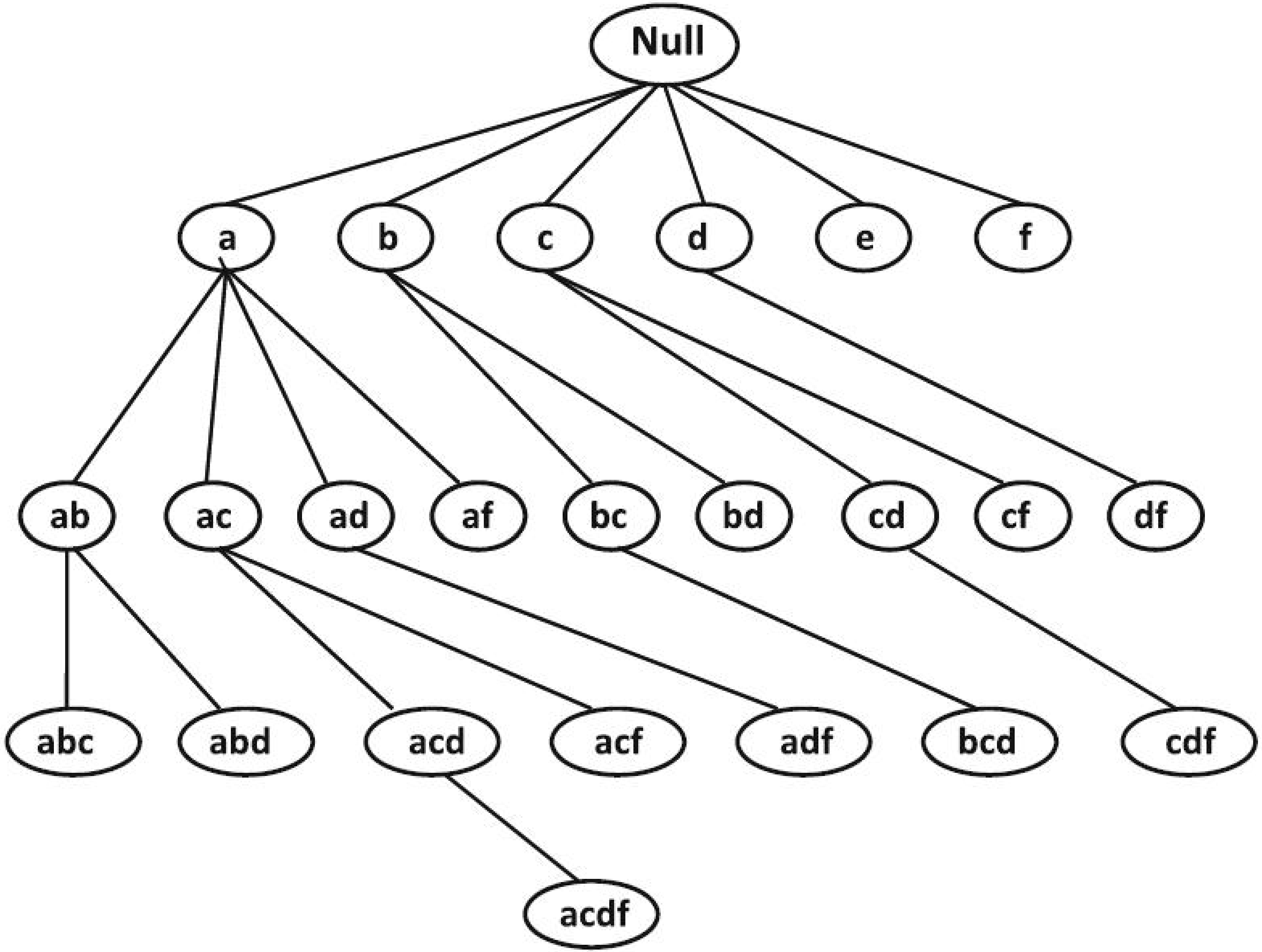


Figure 4.3: The lexicographic or enumeration tree of frequent itemsets

Discussion #4

1. Describe the important concepts in enumeration tree algorithms.
2. The aim of an enumeration tree algorithm is to have a systematic exploration of the candidate patterns in a non-repetitive way.
    - Why is “non-repetitive” an important criterion?
    - Why is “systematic” an important criterion?

Assumption of all these algorithms

- Every data point in a multidimenionsal dataset is:

  independent and identically distributed (i.i.d.)

- In other words, the dataset is generated i.i.d from the same (unknown) distribution

Discussion #5

1. Given a multidimensional dataset, you are consulted as a data mining consultant to explore ways to extract interesting information from this dataset. Explain which algorithm(s) you will use and why.
2. Can we use the data mining algorithms we have studied for a spatial (or graph) dataset? Explain your answer.

Week 4 Summary

- Each data mining task can be solved in different ways using different algorithms.
- Each algorithm follows some problem-solving principle to solve the problem of a data mining task. It is more important to understand the principle than the specific algorithmic details.
- The algorithms we have studied assume that the data is multidimensional.
- To deal with complex data objects, we can still make use of these algorithms as long as we can convert the complex data objects into a multidimensional dataset.

Reflection

- What have you learned this week?
  (Every student shall attempt to answer this question.)